Youssef Miled

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Education

UC Berkeley, *MEng in Operations Research*, California, USA — Aug 2025 – May 2026

- Coursework: Mathematical Programming, Stochastic Optimization for Machine Learning, Reinforcement Learning, Agentic AI.
- **Fung Institute MEng Technical Leadership Award**: Led capstone team to best project award for technical scoping and solution design.

Centrale Lyon, *BSc in General Engineering*, Lyon, France — 2023 – 2025

- Coursework: Applied Mathematics, Computer Science, Machine Learning.

Champollion, *Intensive Preparatory Program for the French Engineering Schools*, Grenoble, France — 2021 – 2023

- Coursework: Algorithms, Data Structures, C, OCaml, Discrete Mathematics.

Professional Experience

Graduate Research Assistant, *Satlyt*, San Francisco, USA — Sep 2025 – May 2026

- Architected an edge-AI satellite telemetry system on Jetson Orin Nano under hard memory/latency budgets; built TF-IDF RAG and tool-calling pipeline (llama.cpp JSON schema) for onboard log summarization and fault diagnosis (62.6 tok/s at 7.81/10 diagnostic accuracy on best configuration).
- Built evaluation infrastructure benchmarking 10 prompt configurations across SLMs (1–4B) via LLM-as-judge; isolated that prompt framing outweighs model size for operational log data and quantified RAG's effect on troubleshooting coverage across models.

ML Research Intern, *CISPA Helmholtz Center for Information Security*, Germany — May 2025 – July 2025

- Developed Cascade Unlearning (manuscript in preparation): a machine-unlearning framework that escalates in-context to LoRA to full fine-tuning as forget-set size grows; introduced Reversed Self-Distillation to align sequentially updated model checkpoints and suppress membership signal leakage across the cascade.
- Audited privacy risk with LiRA membership-inference attacks (TPR@1%FPR) on OLMo 2 1B over SST-2, AGNews, and MIT-Movies; parallelized shadow-model training sweeps across an SLURM cluster with Docker.

Projects

Lean Proof Environments: Reward Design for Kernel-Graded Tasks

- Formalized a 784-line Volpano–Smith–Irvine information-flow security type system in Lean 4 (subtyping, while loops, functions) and built a reinforcement learning environment around it, in which task difficulty is set by how much of a lemma's dependency cone is withheld.
- Diagnosed a reward-design flaw in all-or-nothing kernel grading: performance appeared to collapse with proof depth under binary reward, but regrading the same outputs with per-target partial credit from the kernel recovered substantial progress (depth-4: $0.00 \to 0.46$). Rebuilt the environment around that reward and attached a GRPO learner, with content-addressed parallel grading holding kernel verification to $\sim$6% of a training step.

Type System & Soundness Proof for Information-Flow Noninterference

- Designed and implemented (OCaml) a security-typed type system for an imperative language with expressions, assignments, branches, loops, and functions; extended existing typing rules/semantics to soundly handle function calls.
- Formalized proofs as n-ary proof trees over typing judgment $\Gamma \vdash p : \tau$ and built a type checker that mechanically verifies tree validity; proved the rules enforce noninterference (public-output indistinguishability under public-only input variation), guaranteeing zero private-to-public leakage.

Localizing Memorization in Masked Autoregressive Image Models

- Ported UnitMem neuron-level memorization analysis from SSL encoders to the RAR masked-autoregressive generator (MaskGIT-VQ tokenizer, ImageNet-1k-256), computing per-unit selectivity over intermediate-layer activations under training-time augmentations.
- Traced top-memorizing units back to the individual training images driving their UnitMem scores and compared memorization across layer depth, yielding a per-neuron map of where training data is retained in a generative model.

AgentProbe: Adversarial Agent Evaluation Platform

- Built a distributed four-agent pipeline (Recon, Attack, Evaluator, Reporter) running adaptive multi-turn adversarial scenarios against a target agent in a live sandbox across 5 OWASP-aligned threat families, with a hybrid deterministic-rule + LLM-judge scoring engine and FastAPI/WebSocket streaming evaluation dashboard.

Skills

Languages & Systems	Python, OCaml, C, SQL, JavaScript/TypeScript, React, Next.js, Git, SLURM
Formal Methods / PL	Type systems, soundness proofs, proof trees, information-flow security
Infra & Distributed Systems	Docker, Redis, RQ, PostgreSQL, FastAPI, AWS
AI / Agent Stack	LangGraph, CrewAI, AG2, MCP, RAG, vector databases, tool-calling